REVERSE ENGINEERING REPORT

EMAFORO LED DRIVER — 12V DC

Basis of analysis: supplied product photographs and the provided Reverse Engineering activity document.

Important evidence rule: observations visible in the photographs are stated as facts; component values, ratings, exact part numbers, historical versions, and electrical performance are marked as unverified where they cannot be established from the supplied evidence.

This revised report follows the topic titles and sequence of the supplied Word document. The separate “Day 1 / Day 2 / Day 3” labels are intentionally not used as report headings.

1. Physical Inspection — Studying the Product

Start with the device without opening it. The supplied photographs show a white enclosed LED driver with screw-terminal connections and a 12V DC output marking.

Item	Observation / Answer
Product name/model	EMAFORO LED DRIVER. The label also states “EMMS 5MTS LED” and “OUTPUT DC/12V”. No specific model number is visible.
Manufacturer	EMAFORO (as printed on the front label).
Approximate age/version	Not established from the supplied photographs. No date code or version number is clearly visible.
External dimensions	Exact dimensions cannot be measured reliably from the photographs alone. The enclosure is a small rectangular, wall-mount style module.
Buttons/switches	No external buttons or switches are visible.
LEDs/displays	No user-facing LED indicator or display is visible on the enclosure.
Connectors	Two green 2-position screw-terminal blocks are visible: one at each end of the enclosure.
Power source	The internal PCB clearly contains an AC-to-DC switching power-supply section with a transformer, rectifier components and bulk capacitors. The exact mains input rating is not visible in the supplied photos.
Ports	No USB or data ports are present. The electrical connections are through screw terminals.
Sensors	No sensors are visible or expected from the observed architecture.
Mechanical interfaces	White plastic enclosure with mounting tabs/flanges at the ends; the PCB is enclosed inside the case.
Labels and markings	“EMAFORO”, “LED DRIVER”, “EMMS 5MTS LED”, “ENERGY SAVER”, and “OUTPUT DC/12V” are visible. The output polarity marking is also shown.
Visible screws/seams	The enclosure has a removable cover/seam. The PCB itself uses soldered components; terminal screws are externally accessible.
Ventilation/openings	No obvious ventilation slots are visible on the photographed enclosure.

A. What is this product supposed to do?

It is a mains-powered LED driver/power supply intended to convert electrical input power into a regulated low-voltage DC output for an LED load. The label explicitly identifies the output as DC/12V, so the most defensible conclusion is that it supplies approximately 12 V DC to a compatible LED strip/load.

B. What does the user expect?

The user would expect a compact, reasonably safe and reliable 12 V supply that can continuously power the intended LED strip, maintain an appropriate output voltage, avoid excessive heating, and fit conveniently into the installation. The “ENERGY SAVER” marking also suggests an expectation of efficient operation, although actual efficiency cannot be confirmed from the photographs.

C. What engineering decisions can we already infer?

• A switching power-supply topology was selected rather than a simple linear regulator because the product converts mains power to low-voltage DC.

• A high-frequency transformer provides isolation and enables voltage conversion.

• Bridge-rectifier/diode components and electrolytic capacitors indicate rectification and filtering.

• The output is brought to screw terminals, prioritising simple field wiring for an LED installation.

• The circuitry is enclosed in a plastic housing, providing mechanical protection and reducing user exposure to the electrical circuitry.

2. Product Literature Review

Now move from physical observation to historical research. For this particular unit, the supplied material does not contain manufacturer literature, a model number, a catalogue page, or dated versions. Therefore a genuine Version 1 → Version 2 → Version 3 comparison cannot be established from the evidence supplied.

Parameter	Old version	New version / current unit
Processor	Not applicable / no digital processor identified	No digital MCU identified from the photographs
Battery	Not applicable	No battery
Display	None observed	None observed
Connectivity	No data connectivity	No data connectivity; screw terminals
Charging	Not a battery charger	Not a battery charger; AC-to-DC LED supply
Sensors	None observed	None observed
PCB technology	Unknown	Single PCB with through-hole components visible
Power consumption	Not established	Not established; label says “ENERGY SAVER”
Safety features	Not established	Transformer isolation and enclosed housing are visible design features; protection ratings are not established

Why did the manufacturer change it?

No documented product-generation change is available in the supplied material, so a specific manufacturer change cannot responsibly be claimed. Any statement about a previous version, redesigned circuit, changed transformer, or altered efficiency would require manufacturer literature or additional evidence.

Evidence vs hypothesis

> Evidence: the present PCB contains a transformer, rectifier/diode components, electrolytic capacitors, resistors and a transistor-like switching device, and the enclosure is marked for 12 V DC output. Hypothesis: these parts form an isolated switching AC-to-DC converter suitable for an LED load. The hypothesis should be verified by tracing the PCB and measuring the circuit.

3. Product Analysis — Group Discussion

Purpose

What problem is this product solving?

It solves the problem of supplying a suitable low-voltage DC electrical supply to an LED lighting load from an available mains supply. LEDs generally require controlled low-voltage DC power, while the installation may be connected to mains AC. The driver performs the required conversion and provides a practical connection to the LED load.

User expectations

What would users consider a good product?

- Stable 12 V output suitable for the intended LED strip.
- Reliable operation for long periods.
- Low heat generation and reasonable efficiency.
- Electrical isolation and appropriate protection.
- Compact enclosure and easy installation.
- Secure screw-terminal connections.
- Low cost, because LED drivers are commonly used in quantity.

Trade-offs

What compromises did the designer probably make?

• Cost versus premium protection: a simple through-hole switching design keeps manufacturing and servicing costs low, but may not provide the same sophistication as a high-end industrial supply.

• Size versus thermal performance: a compact enclosure saves space but limits natural cooling area.

• Component cost versus efficiency: the designer must choose inexpensive switching, rectification and filtering components while still meeting the required output.

• Output capability versus transformer size: higher output power requires larger magnetic components and thermal capacity.

• Protection versus simplicity: adding more protection components can improve robustness but increases cost and board area.

> Reverse engineering is therefore not merely “What components are inside?” It is “Why was the product designed this way?”

4. Introduction to Guided Teardown

Only after documentation and research should the device be opened. The supplied photographs provide a before/after-style teardown record showing the enclosure, PCB top and PCB underside.

> The sequence should be: Document → Photograph → Open → Photograph → Identify → Record

Observed teardown sequence

• The external enclosure was first documented.

• The cover was separated from the main housing.

• The PCB was removed/exposed.

• The component side of the PCB was photographed.

• The solder side/tracks of the PCB was photographed.

• The green screw terminals and internal mounting arrangement were documented.

Mechanical observations

• White plastic enclosure with a rectangular internal cavity.

• PCB is mounted inside the enclosure and is accessible after removing the cover.

• Two-position green screw terminals provide the external electrical interfaces.

• The PCB is predominantly populated with through-hole components, with a transformer and electrolytic capacitors occupying significant board area.

• The enclosure provides electrical/mechanical separation between the user and the PCB.

Tools

The source document recommends a precision screwdriver set, plastic pry tools, spudger, tweezers, ESD wrist strap, ESD mat, magnifying glass/microscope, digital camera/phone, multimeter, screw containers and labels/markers. These are appropriate for this teardown.

5. ESD Precautions

Students should understand why ESD matters. A person can carry a static charge that is not noticeable to them but can damage semiconductor devices.

Before touching the PCB:

• Ground yourself.

• Use ESD protection.

• Handle the PCB by its edges.

• Avoid touching IC/transistor pins unnecessarily.

Additional safety for this product

• The PCB is a mains-powered switching supply, so it must never be handled while connected to mains.

• The large electrolytic capacitor can retain dangerous voltage after power is removed.

• Disassembly and electrical testing should be performed by a competent person using appropriate isolation and measurement procedures.

• Do not assume that the output side is safe to touch merely because it is marked 12 V; the isolation and fault condition should be verified before treating it as SELV.

> The source activity specifically requires power to be removed before disassembly and warns that capacitors may retain hazardous energy.

6. Internal Hardware Analysis — Part I

Now create a component inventory. The photographs do not provide enough resolution to establish every exact part number, so the inventory below records visible component types and reference designators where readable.

Ref./Location	Component	Package / form	Marking	Function / notes
T1	High-frequency transformer	Through-hole magnetic component	No readable part number	Provides galvanic isolation and participates in switching voltage conversion.
D1–D7 (visible designators)	Diodes / rectifier devices	Axial through-hole	Individual markings not reliably readable	Rectification, switching or protection depending on circuit position.
Q1/Q2	Transistor / switching devices	Through-hole	Exact markings not readable	Likely switching/control elements in the SMPS primary-side circuitry.
C1/C5 and other electrolytics	Electrolytic capacitors	Radial through-hole	Values not reliably readable	Bulk energy storage and ripple filtering.
R1–R7 and others	Resistors	Axial through-hole	Values not all readable	Biasing, current limiting, sensing/start-up/snubber functions depending on location.
C3/C4 and other capacitors	Capacitors	Through-hole	Values not established	Filtering, timing or EMI/snubbers depending on position.
CN1	Screw terminal	2-position	Green terminal block	One external electrical interface; exact input/output assignment must be traced.
CN2	Screw terminal	2-position	Green terminal block	Other external electrical interface; output is identified as DC/12V on label.
PCB	Single printed circuit board	Through-hole dominated	BMS / 999A markings visible	Carries the complete power-conversion circuit.

7. Component Functional Analysis

This is the most important activity in the module: do not stop at identifying a component. Ask what role it plays in the product.

What role does the main switching section play in this product?

The switching section rapidly switches the rectified input energy through the transformer. This enables the product to achieve voltage conversion and isolation with a much smaller transformer than would be required at mains frequency. The secondary side then rectifies and filters the transferred energy to obtain the low-voltage DC output.

Function Table

Component / block	Electrical role	Functional role	Connected to / relationship
Input terminal	Mains electrical connection	Receives external AC supply	Primary-side rectifier
Rectifier diodes	AC-to-DC rectification	Convert AC to pulsating DC	Bulk capacitor / switching stage

Component / block	Electrical role	Functional role	Connected to / relationship
Bulk electrolytic capacitor	DC energy storage / filtering	Smooths rectified input	Rectifier and primary switching stage
Switching transistor(s)	Electronic switching	Controls energy transfer at high frequency	Transformer / control network
T1 transformer	High-frequency magnetic conversion	Transfers energy while providing isolation	Primary switching stage and secondary rectifier
Secondary diodes	Rectification	Convert transformer secondary waveform to DC	Output filter
Output capacitor(s)	Filtering	Reduce output ripple	12 V output terminals
Resistor network	Bias/sensing/current limiting	Sets operating conditions and protection/control functions	Switching/control nodes
Output terminal	DC electrical connection	Delivers low-voltage power to LED load	LED strip/load

Functional block diagram

> AC MAINS INPUT → RECTIFIER → BULK DC FILTER → HIGH-FREQUENCY SWITCHING → TRANSFORMER → SECONDARY RECTIFIER → OUTPUT FILTER → 12 V DC LED LOAD

The exact control-loop topology cannot be established from the photographs alone because the component markings and underside trace mapping are not sufficiently resolved. A complete schematic should therefore be produced only after continuity tracing and component identification.

The five levels of thinking

• Level 1 — Observation: a white enclosed module, two-position terminals, transformer, diodes, capacitors and resistors are visible.

• Level 2 — Identification: the board can be divided into input, switching, transformer, rectification and output-filter sections; exact semiconductor part numbers remain to be identified.

• Level 3 — Specification: the enclosure label establishes a 12 V DC output, but current/power rating and individual component ratings require further measurement or datasheets.

• Level 4 — Functional understanding: the PCB converts input electrical power through a switching transformer to produce a low-voltage DC output for an LED load.

• Level 5 — Engineering reasoning: the designer appears to have prioritised compact size, low component cost, simple screw-terminal installation and adequate power conversion for LED lighting. This is an inference and should be verified against measured performance and any manufacturer documentation.

> That Level 5 is the real objective of reverse engineering.

Recommended student deliverables

By the end of these three days, every team should produce:

1. External Product Inspection Sheet	Completed above under “1. Physical Inspection — Studying the Product”.
2. Product Version/Literature Comparison	Completed above; historical versions are marked unverified because no literature/model history was supplied.
3. User Needs & Trade-off Analysis	Completed above under “3. Product Analysis — Group Discussion”.
4. Teardown Photo Log	Supplied photographs are included below.
5. Mechanical Disassembly Sequence	Completed above under “4. Introduction to Guided Teardown”.
6. Component Inventory/BOM	Completed above under “6. Internal Hardware Analysis — Part I”.
7. Component Marking & Datasheet Table	Visible markings are recorded; exact datasheet identification remains pending where markings are unreadable.
8. Functional Block Diagram	Completed above under “7. Component Functional Analysis”.

9. Component Function Table	Completed above under “7. Component Functional Analysis”.
10. Final Reverse-Engineering Report	This document.
Reverse Engineering Conclusions	Completed below.

Reverse Engineering Conclusions

What did we learn about the designer's decisions that was not obvious from simply looking at the product?

• The product is not merely a simple 12 V adapter; the PCB implements a switching power-conversion architecture using magnetic isolation, rectification, filtering and switching components.

• The choice of a high-frequency transformer indicates a deliberate compromise between electrical isolation, conversion efficiency, transformer size and cost.

• The through-hole component population and simple screw-terminal interface suggest a design focused on low-cost manufacture and straightforward installation rather than maximum miniaturisation.

• The enclosure protects the user from the PCB and provides a mechanically simple installation form.

• The “ENERGY SAVER” marking indicates that efficiency is part of the product positioning, but this cannot be verified quantitatively without measuring input/output power.

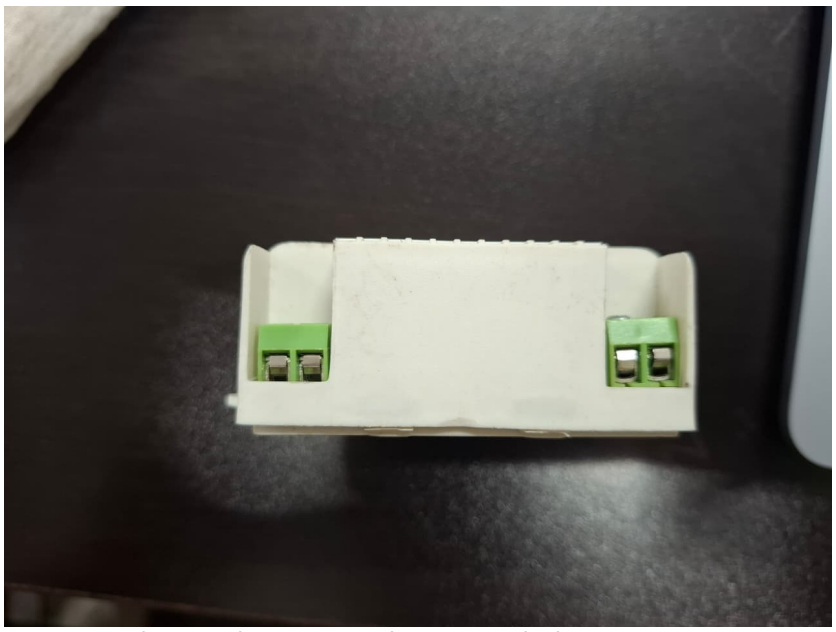
• The exact output current/power capability, switching frequency, regulation method, protection thresholds and component ratings cannot be concluded reliably from the photographs alone.

Overall conclusion: the supplied evidence supports identifying the unit as an enclosed 12 V LED switching driver with an AC-to-DC conversion architecture. Further reverse engineering should focus on reading semiconductor markings, tracing the PCB underside, identifying the controller/switching topology, and measuring the output voltage/current and efficiency under a controlled test setup.

Teardown Photo Log



Photograph 1 — Product label / front identification



Photograph 2 — External screw-terminal arrangement



Photograph 3 — Rear cover / enclosure



Photograph 4 — PCB component side



Photograph 5 — PCB solder / track side